UiT Norges arktiske universitet

Chapter 2 part 2
The Simple Linear Regression Model

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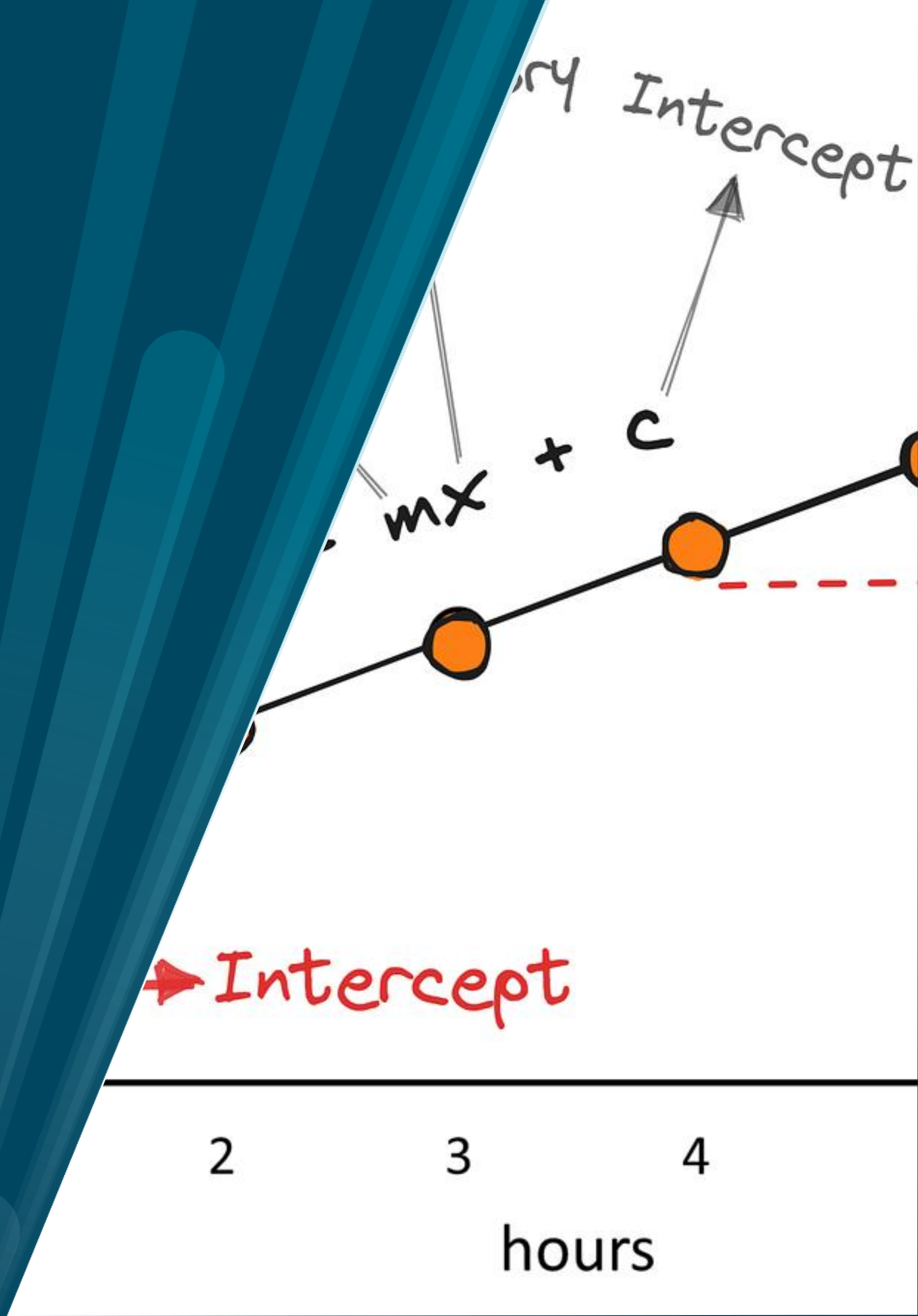

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The simple regression model

$$\underbrace{y_i}_{\text{outcome}} = \underbrace{\beta_1 + \beta_2 \cdot x_i}_{\text{systematic part}} + \underbrace{e_i}_{\text{everything else}}$$

- y_i called dependent variable
  - Because x_i tries to predict it
- x_i called independent variable
- e_i links the observed data and with the precited value on the regression line

Theoretical → Estimated

- When we start estimating parameters change

$$y_i = \beta_1 + \beta_2 \cdot x_i + e_i$$

$$\hat{y}_i = b_1 + b_2 \cdot x_i$$

$$\hat{e}_i = y_i - \hat{y}_i$$

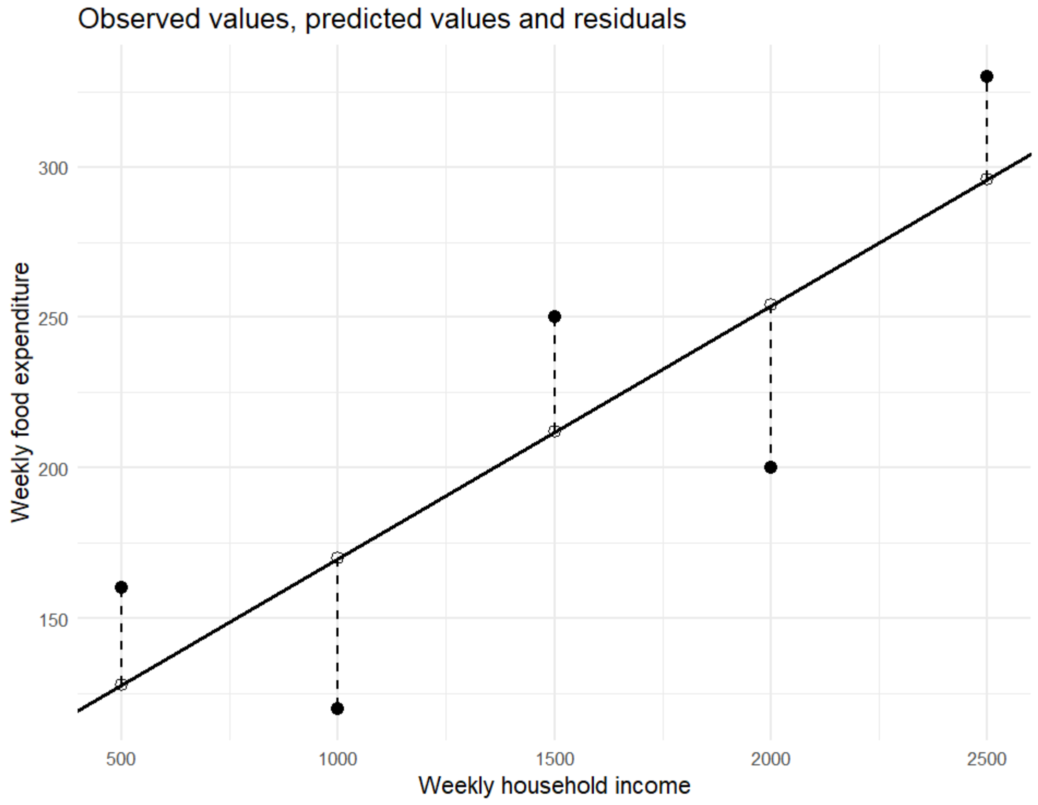
Theoretical / Population	Estimated / Observed	Meaning
y_i	$\hat{y}_i$	Observed outcome / Fitted value
β_1, β_2	b_1, b_2	Population parameters / OLS estimates
e_i	$\hat{e}_i$	Random error / Residual

Lab 1

Minimizing squared errors

$$\widehat{e_i} = y_i - \widehat{y_i}$$

$$SSE = \sum_i (y_i - b_1 - b_2 \cdot x_i)^2$$



$$b_1 = \bar{y} - b_2 \cdot \bar{x}$$

$$b_2 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

The simple regression assumptions

Rule	Meaning	Example / intuition
SR1: Linear regression model $y_i = \beta_1 + \beta_2 x_i + e_i$	The dependent variable is explained by a linear function of x_i plus an error term.	Income may explain part of food expenditure, while e_i captures other influences.
SR2: Zero conditional mean $E(e_i \mid x) = 0$	On average, the error term is zero for every value of x.	For households with a given income, omitted factors should not systematically push food expenditure up or down.
SR3: Constant variance $\mathrm{Var}(e_i \mid x) = \sigma^2$	The variance of the errors is the same for every observation. This is **homoskedasticity**.	The spread of food expenditure around the regression line is similar for low- and high-income households.
SR4: No correlation between errors $\mathrm{Cov}(e_i, e_j \mid x) = 0,\ i \neq j$	The error for one observation is unrelated to the error for another observation.	An unexpectedly high expenditure for household $i$should not imply a high error for household j.
SR5: Variation in x $\mathrm{var}(x) \neq 0$	The sample must contain at least two different values of x.	If every household had exactly the same income, we could not estimate how expenditure changes with income.
SR6: Normal errors (optional) $e_i \mid x \sim N(0, \sigma^2)$	Conditional on x, the errors follow a normal distribution. Mainly useful for exact small-sample inference.	For a given income, most errors are close to zero, with increasingly fewer large positive or negative errors.

Terminology

Concept	**Mean example**	**Example**	**What is it?**
Parameter	μ_x	β_2	An unknown constant in the population.
Estimator	$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$	$b_2 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$	A rule used to estimate the parameter from a random sample. Because the sample is random, the estimator is a random variable.
Estimate	$\bar{x} = 1960.47$	For example, $b_2 = 0.1021$	The numerical value we obtain when the estimator is applied to the observed sample.

Lab 1

Elasticise

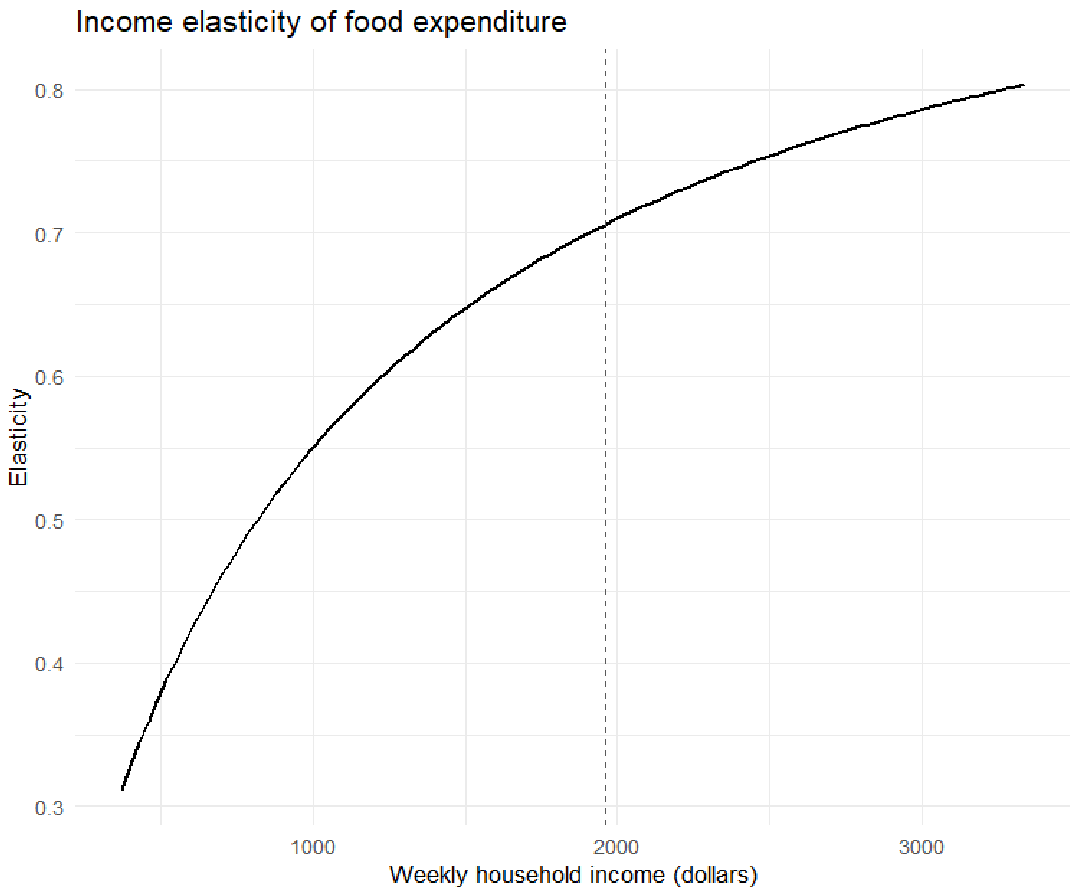
- Using income in dollars changes the estimated slope to 0.1021.
- The fitted values remain identical when the unit conversion is correct.
- For a linear model, elasticity at $(x, \hat{y})$ is $\beta_2 \cdot \frac{x}{\hat{y}}$.
- At the sample means, the food-expenditure elasticity is about 0.706.

Lab 2

Elasticity and income

$$\hat{\varepsilon}(x) = b_2 \frac{x}{\hat{y}} = b_2 \frac{x}{b_1 + b_2 \cdot x_i}$$

- Food expenditure increases with income, but less than proportionally.
- Elasticity increases with income, but at a decreasing rate.
- In our sample, elasticity remains below 1 → food is a necessity.
- At the same time, food expenditure becomes a smaller share of income as income rises.



One sample is only one possible sample

$$\hat{y}_i = 83.416 + 10.209 \cdot x_i$$

If we collected data from another 40 households, would we get exactly the same slope and exactly the same intercept?

One sample, one estimate

- Our sample gives us one estimate of the slope, b_2.
- A different random sample would usually give a different estimate.
- This is called **sampling variability**.

$$\hat{y}_i = b_1 + b_2 \cdot x_i$$

Same estimator, but different sample leads to different estimate

- What would happen if we could repeat the study many times?

Suppose we knew the truth

Imagine that we are **psychic**

We know the true population model:

$$y_i = \beta_1 + \beta_2 \cdot x_i + e_i$$

We know the true population parameters:

$$\beta_1 = 80$$
$$\beta_2 = 0.1$$
$$e_i \sim N(0, 50^2)$$

Some researchers conduct a survey in this population.

- They collect a random sample of **40 households** and estimate a regression from the sample.
- What estimates will they get?
- Will they estimate exactly
- And what happens if another researcher collects **another sample of 40 households**?
- We can recreate this experiment on a computer and repeat it many times.

This is called a Monte Carlo simulation.

Lab 3

Monte Carlo simulation: the idea

We know the true population model:

$$y_i = 80 + 0.10x_i + e_i$$

with

$$\beta_1 = 80$$
$$\beta_2 = 0.10$$
$$\sigma = 50$$
$$e_i \sim N(0, 50^2)$$

Then we repeat the same experiment many times:

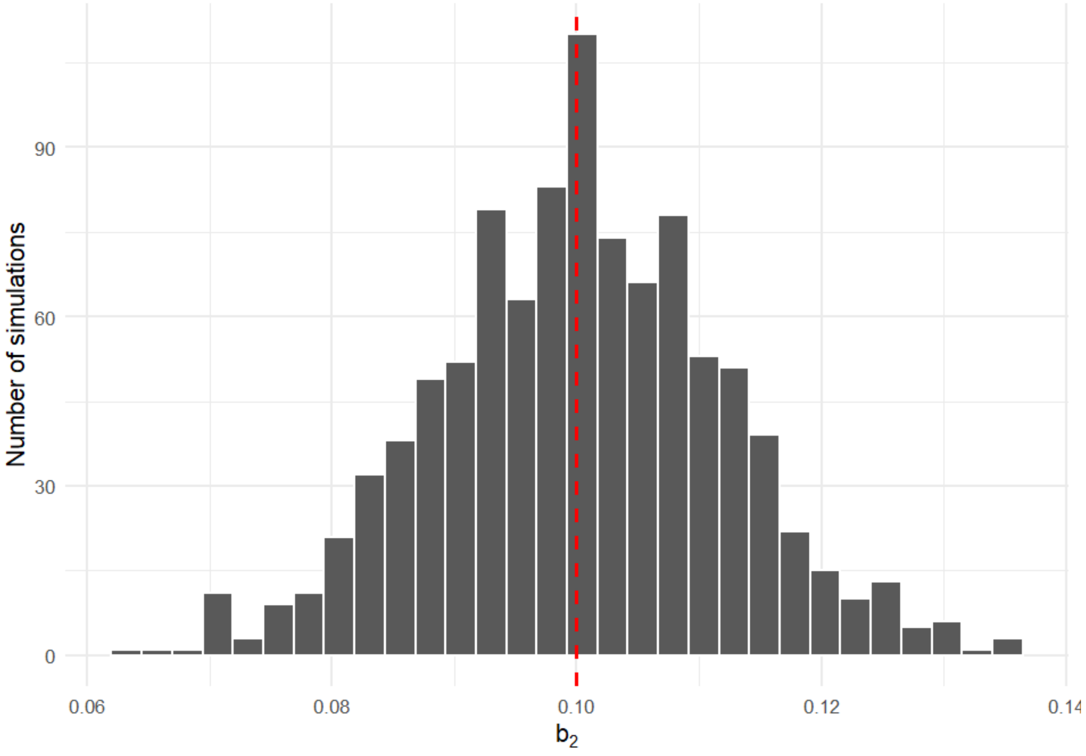
1. Keep the income values x_i fixed.
2. Draw a new random error e_i for each household.
3. Generate a new food expenditure value
4. Estimate the regression and record b_1 and b_2.
5. Repeat the process many times.

What did the simulation show?

True slope: $\beta_2 = 0.10$

Sampling distribution of the OLS slope estimator

1,000 samples from the same population model



Repeated samples produce different regression lines

20 estimated regressions and the true population relationship

The OLS estimator is unbiased

$$E(b_2|x) = \beta_2$$

- The sampling distribution is centered on the true parameter.
- Across many repeated samples, the average estimate approaches β_2.
- Unbiasedness is a property of the **estimator**, not of one particular estimate.

Unbiased does not mean precise

Unbiasedness
$E(b_2|x) = \beta_2$

Where is the sampling distribution centered?

- Centered on the true parameter
- No systematic over- or underestimation

Precision
$var(b_2|x)$

How much does the estimator vary across samples?

- Small variance → estimates stay close together
- Large variance → estimates can vary substantially

An estimator can be unbiased and still be imprecise.

Lab 4

The variance of the slope estimator

$$var(b_2|x) = \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}$$

More noise $\longrightarrow$ lower precision

$$\sigma^2 \uparrow \Rightarrow var(b_2|x) \uparrow$$

More variation in x $\longrightarrow$ higher precision

$$\sum_i (x_i - \bar{x})^2 \uparrow \Rightarrow var(b_2|x) \downarrow$$

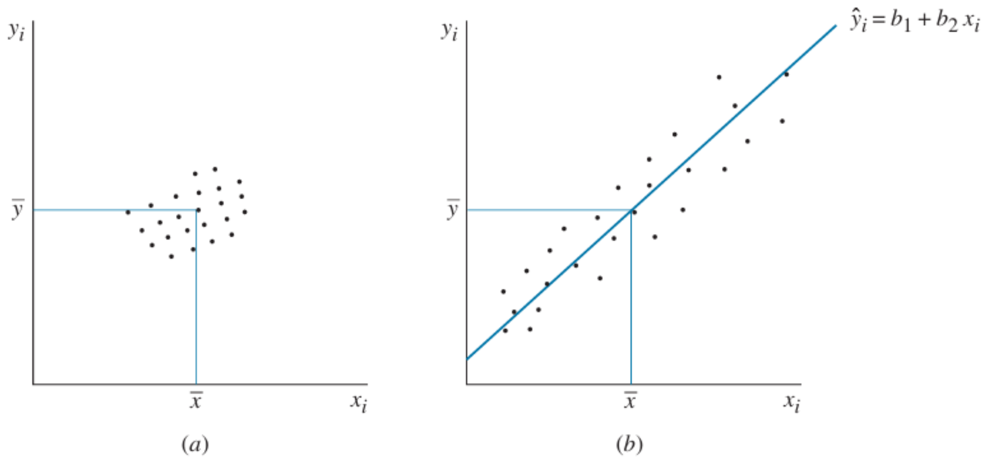


FIGURE 2.11 The influence of variation in the explanatory variable x on precision of estimation: (a) low x variation, low precision: (b) high x variation, high precision.

Precision improves when the signal is easier to distinguish from the noise.

Lab 5

What about the intercept?

- More error variation → less precise b_1
- More observations and more variation in x → more precise b_1
- The intercept is harder to estimate precisely when the observed x-values are far from x=0

$$var(b_1|x) = \sigma^2 \cdot \frac{\sum_i (x_i)^2}{N \cdot \sum_i (x_i - \bar{x})^2}$$

The Gauss–Markov Theorem

- Under assumptions **SR1–SR5**:

OLS is BLUE

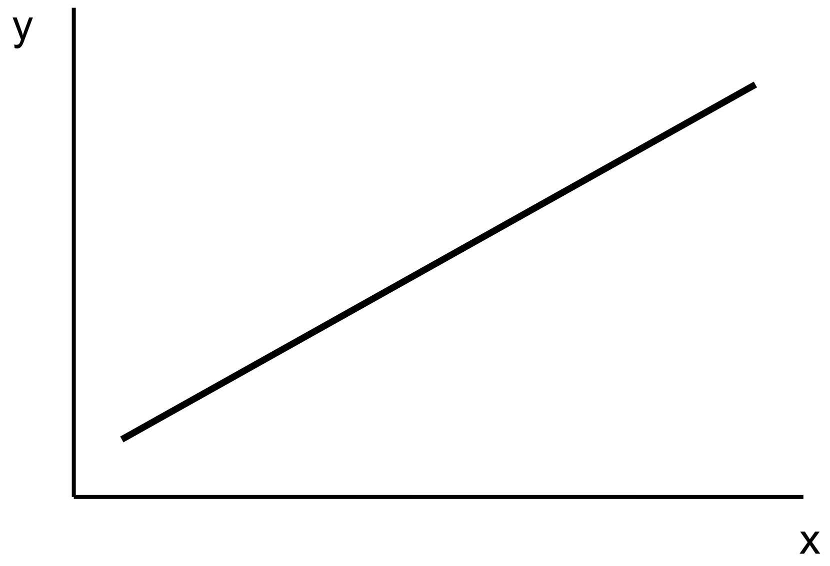
- **Best**
- **Linear**
- **Unbiased**
- **Estimator**

- **Linear** — $b_1$and b_2 are linear estimators.
- **Unbiased** — $E(b_1 \mid X) = \beta_1$and $E(b_2 \mid X) = \beta_2$.
- **Best** — OLS has the **smallest variance** among all linear unbiased estimators.

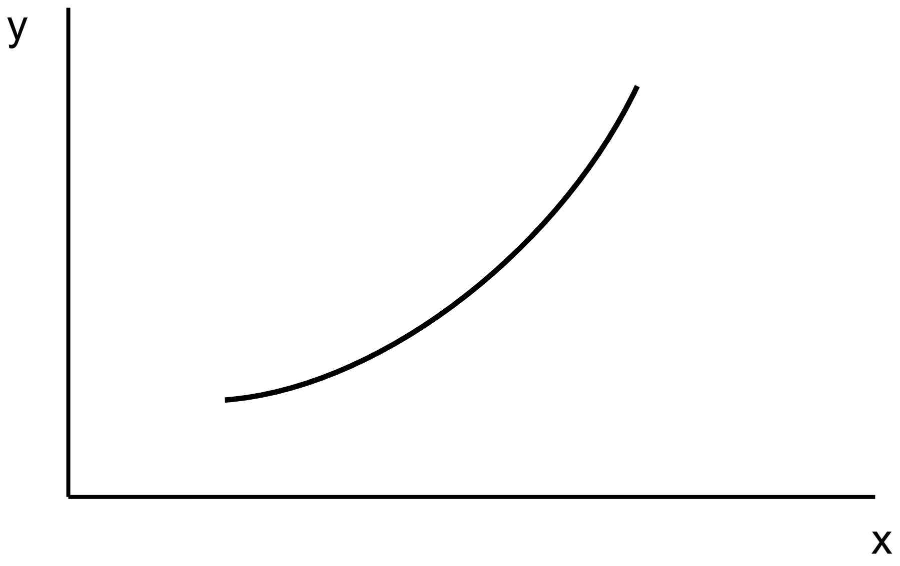
Relationships do not have to be straight lines

Economic relationships are often nonlinear. We can still use the simple regression framework by transforming x and/or y.

Linear



Nonlinear



Linear regression means linear in the parameters — not necessarily a straight line in the original variables.

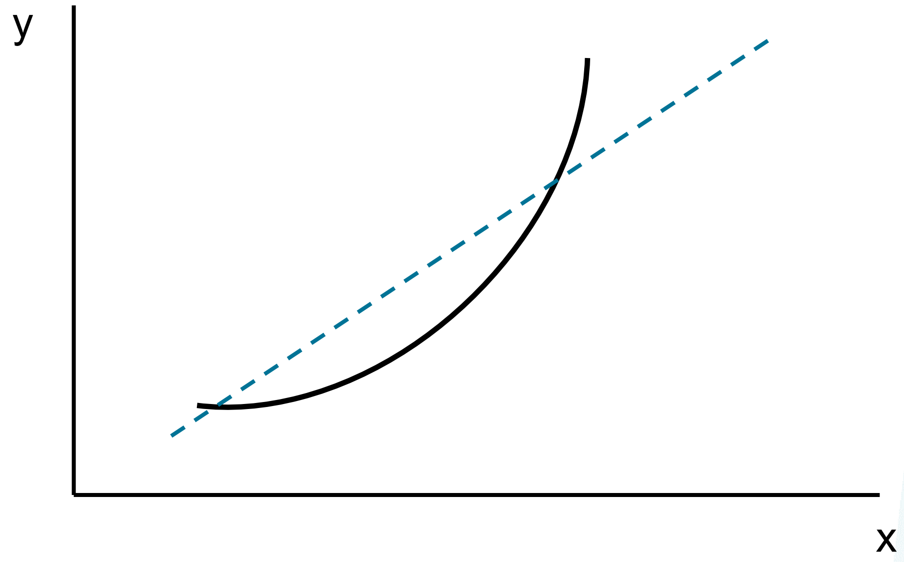
A quadratic relationship

$$y = \beta_1 + \beta_2 \cdot x^2$$

The marginal effect is no longer constant:

$$\frac{dy}{dx} = 2\beta_2 x$$

- The effect of x depends on the level of x.
- The slope changes along the curve

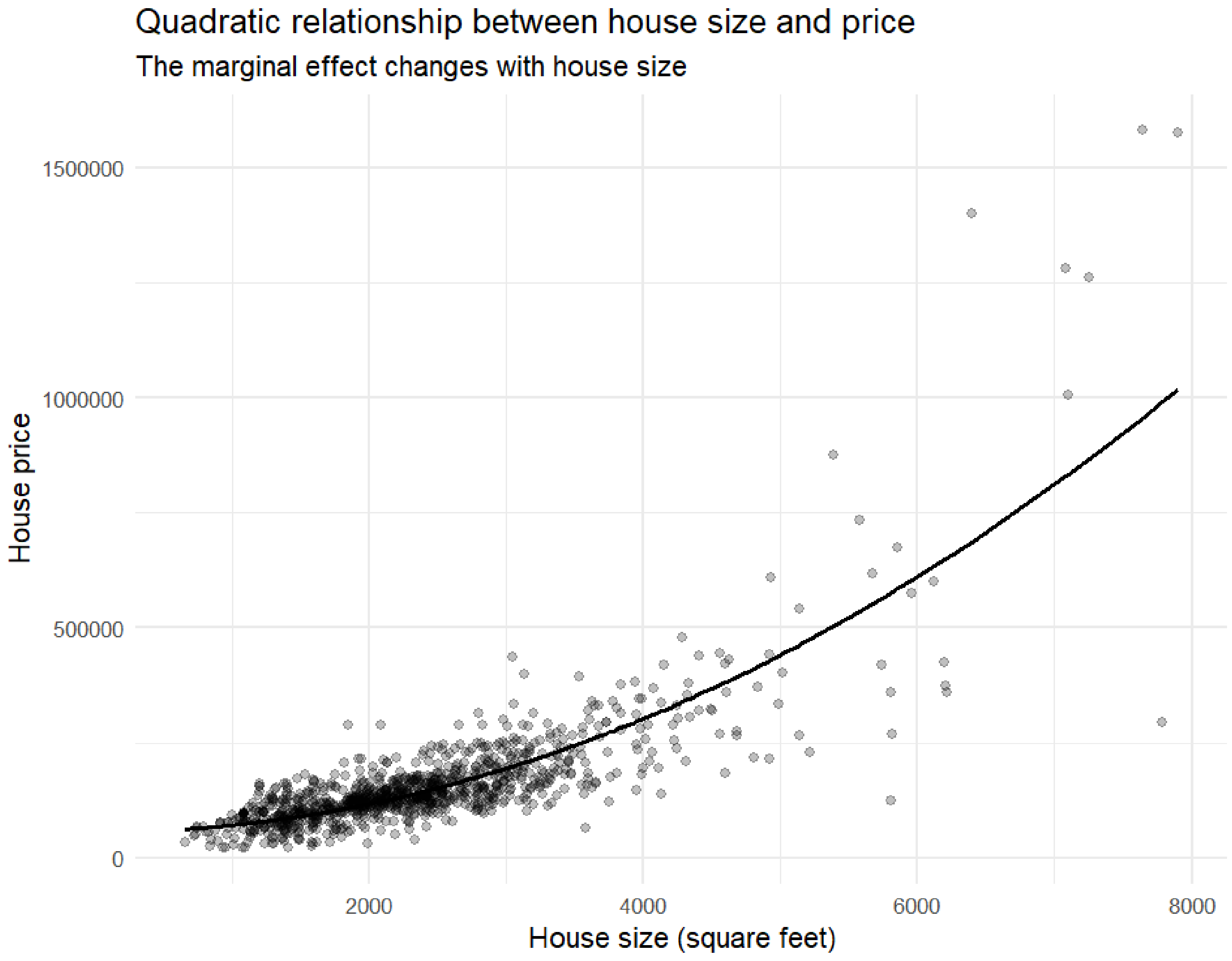


Perdictions

- $\frac{dy}{dx} = 2\beta_2 x = 2 \cdot 0.0154 \cdot x$

Marginal increase

- $x = 2000$
  - $\frac{dy}{dx} = 2 \cdot 0.0154 \cdot 2000 = 61$
- $x = 4000$
  - $\frac{dy}{dx} = 2 \cdot 0.0154 \cdot 4000 = 123$
- $x = 6000$
  - $\frac{dy}{dx} = 2 \cdot 0.0154 \cdot 6000 = 180$



A log-linear relationship

Lab 7

$$\ln(y) = \beta_1 + \beta_2 x$$

Example: $\beta_2 = 0.004$

≈ 0.4% increase in y for a one-unit increase in x